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SELF POWERED PIEZOELECTRIC ENERGY HARVESTER

Problem Statement

A compact, self-powered, microcontroller-free synchronized switching interface circuit (SP-SSHI) using piezoelectric ceramics is developed to harvest irregular and low-frequency mechanical energy from human footsteps. Unlike conventional rectifier-based circuits, this SP-SSHI maximizes harvested energy, reduces losses from impedance mismatch, and stabilizes the output voltage while operating solely on the captured energy. The circuit eliminates the need for batteries, external power, or complex control electronics, enabling efficient, autonomous energy storage and delivery for long-lasting, cost-effective applications in sustainable wearable and sensor systems

Approach / Design / Methodology

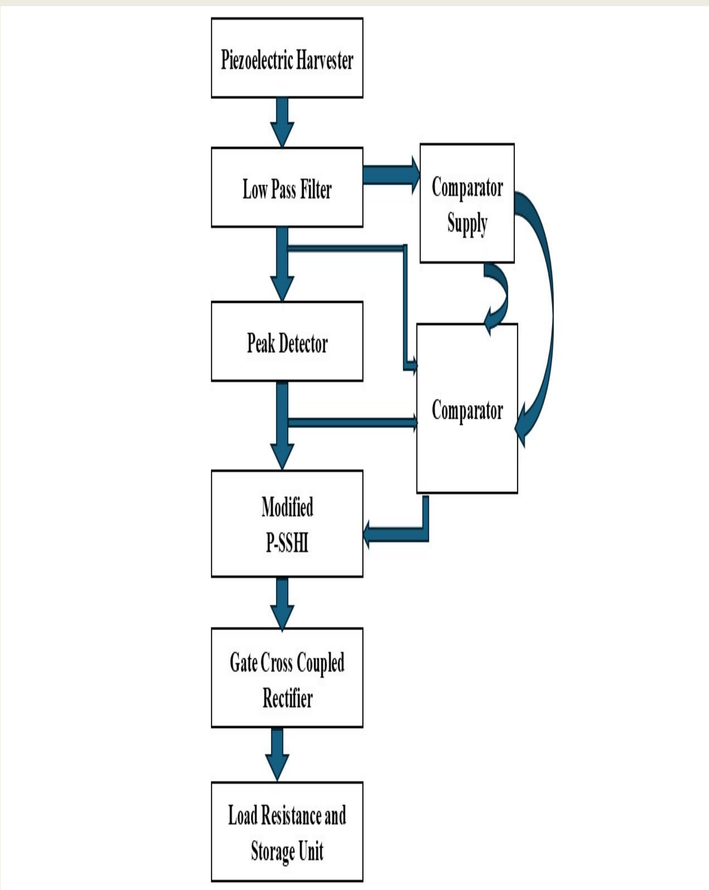

FIG 1 – BLOCK DIAGRAM

- An efficiency-driven architecture was selected to improve power extraction from low-amplitude vibration sources instead of relying on conventional diode rectification.
- A synchronized energy transfer strategy was adopted to ensure switching occurs at optimal waveform instants, reducing energy loss during charge extraction.
- The power processing stage was designed to minimize conduction and switching losses using active rectification techniques.
- The system was structured to enhance charge transfer from the piezoelectric element through controlled voltage inversion.
- Energy accumulation was optimized to convert intermittent vibration inputs into a stable and usable DC output.
- A fully hardware-based implementation was chosen to achieve low power consumption, compactness, and self-powered operation.

Circuit Diagram /Algorithm

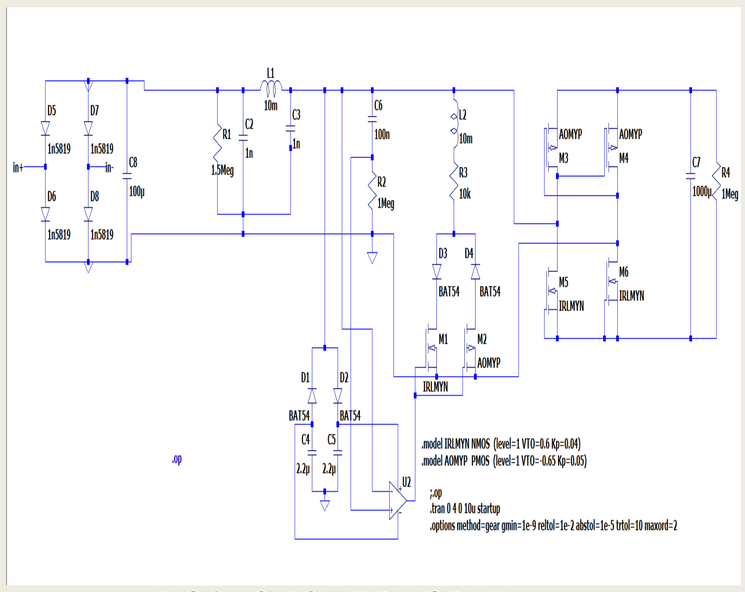

FIG 2 – CIRCUIT DIAGRAM

Simulation Results

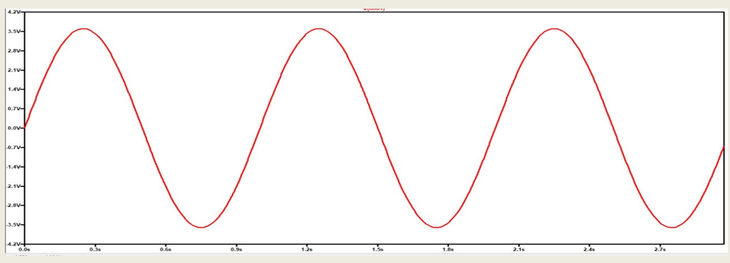

FIG 3 - INPUT SIGNAL TO CIRCUIT

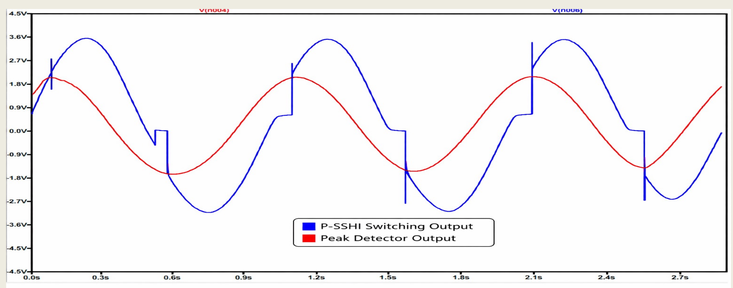

FIG 4 - PEAK DETECTOR AND P-SSHI SWITCHING OUTPUT

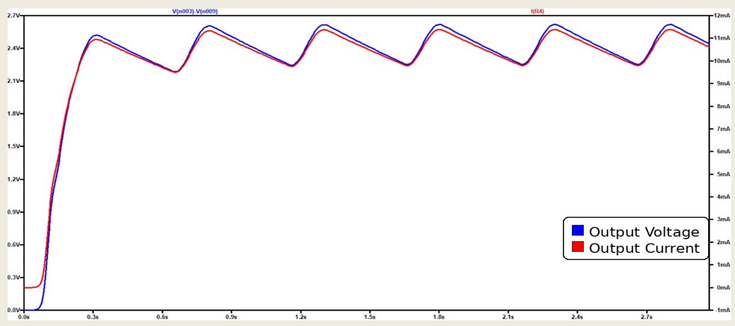

FIG 5 - OUTPUT VOLTAGE AND CURRENT

Working Model Picture

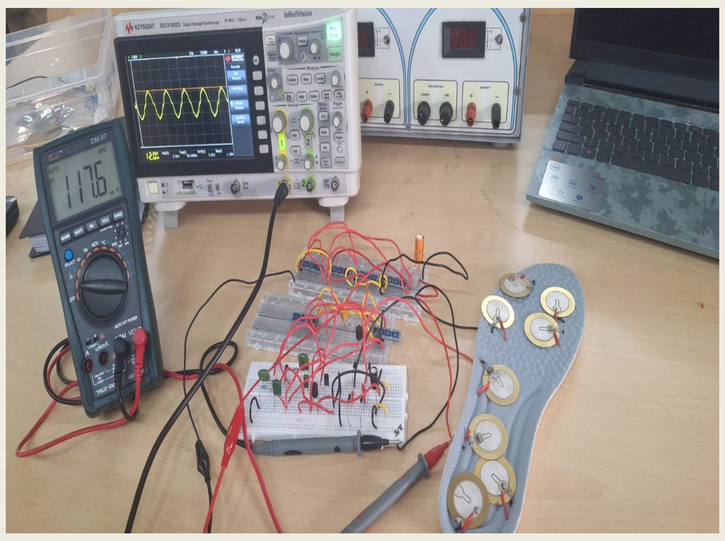

FIG 6 – WORKING MODEL OF THE CIRCUIT

PCB Layout

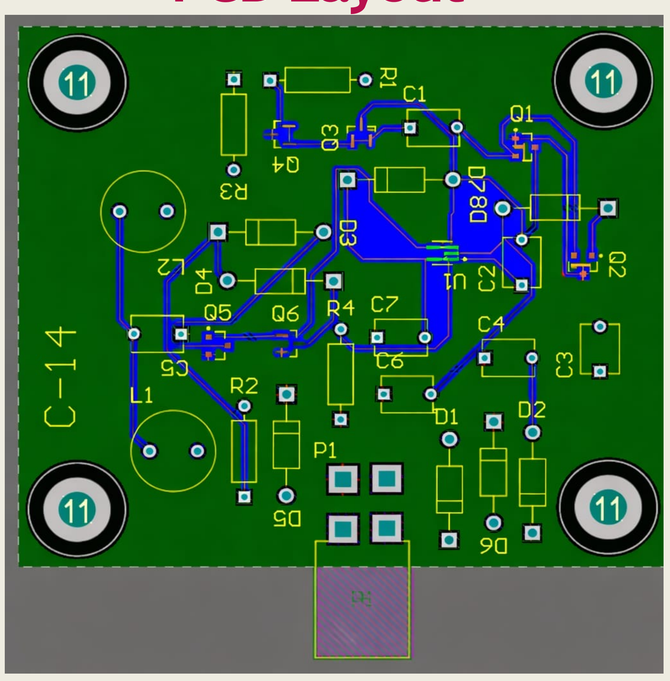

FIG 7 – PCB LAYOUT OF THE CIRCUIT

Results

•The piezoelectric disc generated an AC output when pressed or tapped, producing a sinusoidal waveform on the DSO.
•The output voltage varied with applied stress, ranging approximately from 60 mV to 200 mV, with each tap producing a clear peak.
•Repeated tapping charged the storage capacitor, and the output remained stable when no stress was applied.

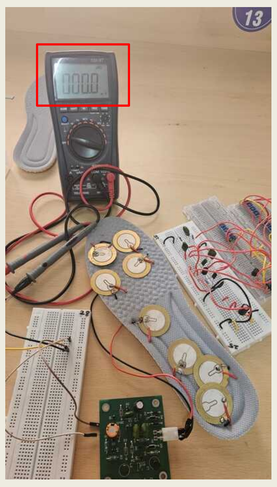

FIG 8 – BEFORE TAPPING

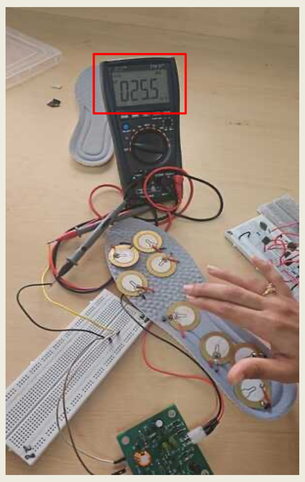

FIG 9 – AFTER TAPPING

References

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